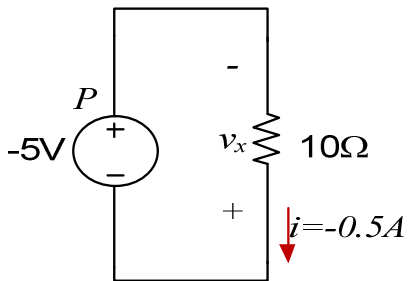


EBN 111 Klastoets / Class Test # 1 (21 Feb 2014)**15 Marks, 30 minutes**

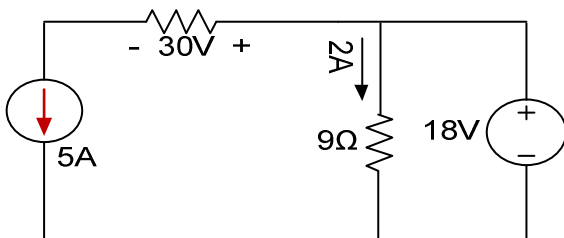
1. Hoeveel energie word in 'n 12V kar battery van 60Ah gestoor? / *How much energy is stored in a 12 V car battery rated 60Ah?* (1)

2. Die lading deur 'n element val van 10mC tot 2mC tussen tyd 0ms en 5ms. Wat is die stroom deur element by 3ms? / *The charge through an element drops from 10mC to 2mC between time 0ms and 5ms. What is the current through the element at 3ms?* (1)

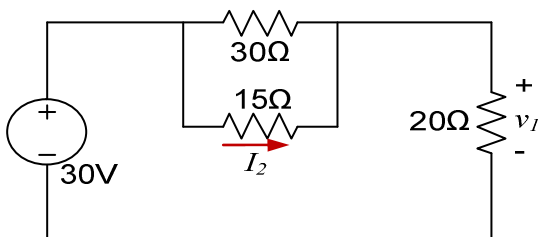
3. Bepaal die spanning v_x oor die resistor, asook die drywing P van die spanningsbron. / *Determine the voltage v_x across the resistor and the power (P) associated with voltage source.* (2)



4. Bepaal die drywing geassiosieer met die 5A stroombron. / *Calculate the power associated with 5A current source.* (2)

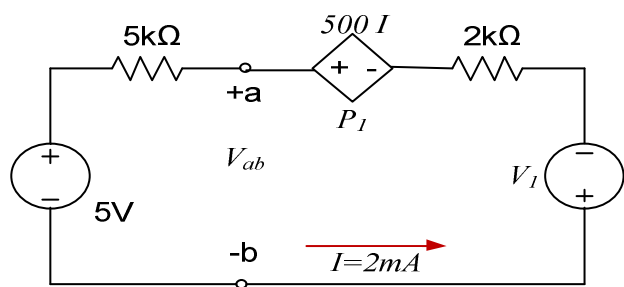


5. Bepaal v_1 en I_2 . / *Calculate v_1 and I_2 .* (2)



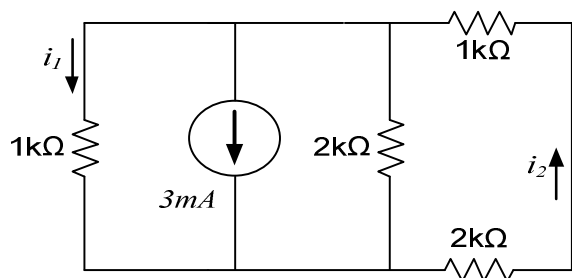
6. Bepaal V_{ab} en V_I . / Determine V_{ab} and V_I ?

(2)



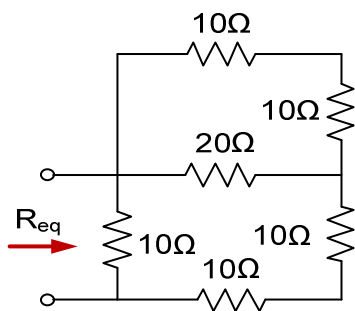
7. Bepaal i_1 and i_2 . / Find i_1 and i_2 .

(2)



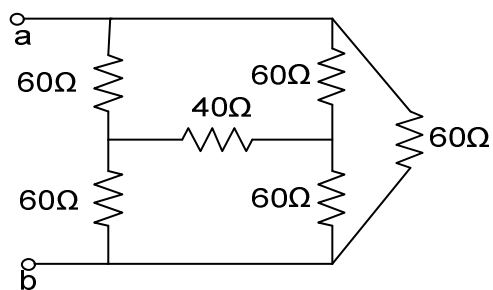
8. Bepaal die ekwivalente weerstand R_{eq} . / Calculate the equivalent resistance R_{eq} .

(1)



9. Bepaal die ekwivalente weerstand R_{ab} tussen terminale a en b . / Find equivalent resistance R_{ab} across nodes a and b .

(2)



EBN111 Klastoets / Class Test # 1 (21 Feb 2014)
ANTWOORDBLAD / ANSWER SHEET

Student besonderhede / <i>Student details</i>									
Van: <i>Surname:</i>					Voorletters <i>Initials:</i>				
Graadkursus: <i>Degree Course:</i>					Studentenommer: <i>Student number:</i>				

MARKS	
TOTAAL/TOTAL	

VRAAG – QUESTION 1	
1	$w =$
2	$I =$
3	$v_x =$ $P =$
4	$P_{5A} =$
5	$v_I =$ $I_2 =$
6	$V_{ab} =$ $V_I =$
7	$i_I =$ $i_2 =$
8	$R_{eq} =$
9	$R_{ab} =$